

ON CONFIDENCE SETS FOR THE RATIO OF TWO NORMAL MEANS

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1. Summary

Fieller's (1940) technique for obtaining a confidence set (c.s.) for the ratio of two normal means is based on a confidence statement about a t-statistic involving a linear combination of estimates of the unknown means. This statement is inverted to yield a quadratic equation, the roots of which are the limits of the desired c.s. The c.s. may be a finite interval, the complement of a finite interval or the real line. The purpose of this paper is to study the probability of obtaining each type of c.s. as well as the probability of coverage in each case. Numerical values of these probabilities are presented.

2. Introduction

Let U_1, U_2 be jointly normally distributed with means μ_1, μ_2 and covariance matrix $\sigma^2(v_{ij})$, $i, j=1, 2$ where the v_{ij} are controllable constants with $v_{11}v_{22}-v_{12}^2 > 0, v_{11} > 0, v_{22} > 0$ and where μ_1, μ_2 and σ^2 are unknown parameters. Let S^2 be an estimate of σ^2 , independent of U_1 and U_2 , such that $S^2 \sim \sigma^2 \chi^2(f)/f$.

Assume $\mu_2 \neq 0$ and define θ by

$$\mu_1 + \theta \mu_2 = 0, \text{ i.e. } \theta = -\mu_1/\mu_2.$$

The problem is to find a c.s. for θ . This problem has been discussed with respect to various applications by Bennett (1957), Box (1954), Creasy (1954), Fieller (1940) and Kastenbaum (1959), and generalizations and extensions thereof by Bennett (1959, 1963), Box and Hunter (1954), Cox (1967), Elston (1969), Halperin (1967), Laubscher (1968) and Robison (1964). A different approach to the regression applications, namely to treat the dependent variable formally as the independent variable and vice versa, is discussed by Berkson (1969), Halperin (1970), Krutchkoff (1967, 1969), Lamborn (1970) and Williams (1969). The Bayesian approach is discussed by Dunsmore (1968) and Hoadley (1970). An important recent contribution is that of Scheffé (1970) who studies the problems of multiple testing and multiple estimation, and proposes methods of avoiding "improper confidence sets" of the form $(-\infty, \infty)$.

Some regression applications of Fieller's method to be found in the literature are confidence sets for the value of the independent variable at which a regression line reaches a given value, at which two regression lines intersect or at which a quadratic regression reaches a maximum or minimum. It can also be used to find a confidence set for the horizontal difference between two parallel regression lines.

3. Confidence sets for θ

The following method was proposed by Fieller (1940) and studied further by

Creasy (1954). A different representation, in terms of t-statistics, is used here. Define $T_\theta = (U_1 + \theta U_2) / S(v_{11} + 2v_{12}\theta + v_{22}\theta^2)^{1/2}$, then $T_\theta \sim$ Student's t with f d.f. Let $t = t_{f/2}^\gamma$ be a critical value of Student's t, i.e.

$$1 - \gamma = P \{ |T_\theta| \leq t \} = P \{ Q(\theta) \leq 0 \},$$

where

$$Q(\theta) = (U_1^2 - S^2 t^2 v_{11}) + 2(U_1 U_2 - S^2 t^2 v_{12})\theta + (U_2^2 - S^2 t^2 v_{22})\theta^2.$$

The set of all h for which $Q(h) \leq 0$ is a c.s. for θ with confidence $1 - \gamma$.

The discriminant D of the quadratic expression $Q(\theta)$ is given by

$$D/4 = vS^4(t_1^2 + T_2^2 - t^2)$$

where

$$T_1 = U_2/v_{22}S, T_2 = v_{22}^{1/2}(v_{12}U_2/v_{22} - U_1)/v^{1/2}S.$$

T_1, T_2 have a bivariate non-central t-distribution with independent numerators and non-centrality parameters (n.c.p.)

$$\delta_1 = \mu_2/v_{22}^{1/2}\sigma, \delta_2 = v_{22}^{1/2}\mu_2(v_{12}/v_{22} - \theta)/v^{1/2}\sigma,$$

while $F = \frac{1}{2}T_1^2 + \frac{1}{2}T_2^2$ has a non-central F-distribution with 2 and f d.f. and n.c.p.

$$\lambda = \delta_1^2 + \delta_2^2 = \mu_2^2(v_{11} - 2v_{12}\theta + v_{22}\theta^2)/v\sigma^2.$$

The coefficient of θ^2 in $Q(\theta)$ is $v_{22}S(T_1^2 - t^2)$.

Let H_1, H_2 be the solutions of $Q(h) = 0$. There are three possibilities:

- $T_1^2 + T_2^2 \leq t^2$ which implies $T_1^2 \leq t^2$: The roots H_1, H_2 are complex and $Q(h) \leq 0$ for all h . Thus the c.s. for θ is $(-\infty, \infty)$.
- $T_1^2 + T_2^2 > t^2$ and $T_1^2 \leq t^2$: The roots H_1, H_2 are real and $Q(h) > 0$ for all $h \in (H_1, H_2)$. Thus the c.s. for θ is $(H_1, H_2)^c$.
- $T_1^2 > t^2$ which implies $T_1^2 + T_2^2 > t^2$: The roots H_1, H_2 are real and $Q(h) \leq 0$ for all $h \in (H_1, H_2)$. The c.s. for θ is thus (H_1, H_2) .

The relationship between T_θ and T_1, T_2 is

$$T_\theta = \rho T_1 + (1 - \rho^2)^{1/2} T_2$$

where $\rho = \delta_2/\lambda^{1/2}$ is the correlation coefficient between the numerators of T_θ and T_1 .

4. Probabilities associated with the types of confidence sets

Define the probabilities $P_1, \dots, P_5$ as follows:

$$P_1 = P \{ T_1^2 + T_1^2 \leq t^2 \} ,$$

$$P_2 = P \{ T_1^2 > t^2, T_\theta^2 \leq t^2 \} ,$$

$$P_3 = P \{ T_1^2 > t^2, T_\theta^2 > t^2 \} ,$$

$$P_4 = P \{ T_1^2 \leq t^2, T_1^2 + T_2^2 > t^2, T_\theta^2 \leq t^2 \} ,$$

$$P_5 = P \{ T_1^2 \leq t^2, T_1^2 + T_2^2 > t^2, T_\theta^2 > t^2 \} .$$

The significance of $P_1, \dots, P_5$ may be summarized as follows:

TABLE 1
Probabilities of obtaining the three types of confidence sets
and coverage probabilities

	Form of the confidence set C			Total
	$C = (-\infty, \infty)$	$C = (H_1, H_2)$	$C = (H_1, H_2)^c$	
$\theta \in C$	P_1	P_2	P_4	$1 - \gamma$
$\theta \notin C$	0	P_3	P_5	γ

Of special importance are $P_2/(P_2 + P_3)$ and $P_4/(P_4 + P_5)$, the conditional coverage probabilities given that the c.s. is a finite interval and the complement of a finite interval respectively.

It follows from Tang (1938) that P_1 can be written

$$P_1 = P \{ F \leq \frac{1}{2} t^2 \} = \sum_{i=0}^{\infty} \exp(-\frac{1}{2} \lambda) (\frac{1}{2} \lambda)^i i!^{-1} I_{t^2/(f+t^2)}(i+1, \frac{1}{2} f)$$

where $I_x(p, q)$ is the incomplete Beta function ratio, while

$$P_1 + P_4 + P_5 = P(T_1^2 \leq t^2) = \sum_{i=0}^{\infty} \exp(-\frac{1}{2} \delta_1^2) (\frac{1}{2} \delta_1^2)^i i!^{-1} I_{t^2/(f+t^2)}(i+\frac{1}{2}, \frac{1}{2} f),$$

and $P_1 + P_2 + P_4 = P(T_\theta^2 \leq t^2) = 1 - \gamma$.

Furthermore

$$P_1 + P_4 = P \{ T_1^2 \leq t^2, T_\theta^2 \leq t^2 \} = \int_{-t}^t \int_{-t}^t p(t_1, t_\theta) dt_1 dt_\theta$$

where $p(t_1, t_\theta)$ is the joint density of T_1, T_θ , namely a non-central bivariate t-density which may be written, as a special case of the multivariate t-distribution in Kshirsagar (1961), provided $\rho^2 \neq 1$, i.e. $\delta_1 \neq 0$,

$$p(t_1, t_\theta) = \pi^{-1} (1 - \rho^2)^{-1/2} \Gamma(1/2 f)^{-1} \exp \left\{ -\frac{1}{2} \delta_1^2 / (1 - \rho^2) \right\} \\ \sum_{i=0}^{\infty} 2^{1/2 i} \delta_1^i (t_1 - \rho t_\theta)^i \Gamma(1/2 f + 1/2 i + 1) f^{-1/2 i - 1} i!^{-1} (1 - \rho^2)^{-i} \\ \left\{ 1 + (t_1^2 - 2 \rho t_1 t_\theta + t_\theta^2) / f (1 - \rho^2) \right\}^{-1/2 f - 1/2 i - 1}.$$

By integration and transformation to polar coordinates it follows that

$$P_1 + P_4 = \pi^{-1/2} \exp \left\{ -\frac{1}{2} \delta_1^2 / (1 - \rho^2) \right\} \sum_{i=0}^{\infty} \Gamma(i + 1/2)^{-1} 2^{-i} \\ \pi^{1/2 + \phi(\rho)} \int_{\phi(\rho)}^{\phi(\rho)} \delta_1^{2i} \left\{ [(1 - \rho^2)^{-1/2} \cos x]^{2i} + [\sin x - \rho (1 - \rho^2)^{-1/2} \cos x]^{2i} \right\} \\ I_{t^2/(t^2 + f \sin^2 x)}^{(i+1, 1/2 f)} dx$$

where $\phi(\rho) = \arctan \{ (1 + \rho) / (1 - \rho) \}^{1/2}$.

A similar expression is obtained for

$$P_3 = P \{ T_1^2 > t^2, T_\theta^2 > t^2 \}$$

where $I_{t^2/(t^2 + f \sin^2 x)}^{(i+1, 1/2 f)}$ is replaced by

$$1 - I_{t^2/(t^2 + f \cos^2 x)}^{(i+1, 1/2 f)}.$$

From the foregoing expressions $P_1, \dots, P_5$ may be computed for given values of f, γ, δ_1 and δ_2 . A number of these values are summarized in tables 2 through 5.

TABLE 2

The effect of increase in γ with ρ constant $(f = 10; \rho = 0,3162; \gamma = 0,05)$

δ_1	δ_2	λ	P_1	P_2	P_3	$\frac{P_2}{P_2+P_3}$	P_4	P_5	$\frac{P_4}{P_4+P_5}$
,3	,1	,1	,8557	,0497	,0088	,8498	,0446	,0412	,5195
,6	,2	,4	,8226	,0738	,0108	,8724	,0536	,0392	,5774
,9	,3	,9	,7680	,1152	,0139	,8923	,0667	,0361	,6490
1,2	,4	1,6	,6937	,1747	,0178	,9074	,0815	,0322	,7170
1,8	,6	3,6	,5030	,3433	,0269	,9274	,1037	,0231	,8177
2,4	,8	6,4	,3021	,5464	,0355	,9389	,1015	,0145	,8753
3,0	1,0	10,0	,1456	,7291	,0423	,9452	,0753	,0077	,9071

The two conditional coverage probabilities increase as λ increases.

TABLE 3

The effect of increase in δ_2 with δ_1 constant $(f = 20; \delta_1 = 3; \gamma = 0,05)$

λ	ρ	P_1	P_2	P_3	$\frac{P_2}{P_2+P_3}$	P_4	P_5	$\frac{P_4}{P_4+P_5}$
9	0	,1389	,7707	,0434	,9467	,0403	,0066	,8598
10	,3162	,1088	,7730	,0412	,9495	,0682	,0088	,8852
12	,5	,0657	,7762	,0379	,9534	,1081	,0121	,8995
15	,6325	,0298	,7792	,0350	,9571	,1410	,0150	,9036
18	,7071	,0131	,7810	,0331	,9593	,1559	,0169	,9023
21	,7559	,0056	,7823	,0319	,9608	,1622	,0181	,8996
24	,7906	,0023	,7831	,0310	,9619	,1645	,0190	,8966
27	,8165	,0009	,7838	,0304	,9627	,1653	,0196	,8938
30	,8367	,0004	,7843	,0299	,9633	,1653	,0201	,8914

 $P_2/(P_2 + P_3)$ increases with δ_2 . $P_4/(P_4 + P_5)$ increases and then decreases.

TABLE 4

The effect of change in δ_1 and δ_2 for fixed λ $(f = 10; \lambda = 25; \gamma = 0.05; P_1 = 0.0042)$

δ_1	δ_2	ρ	P_2	P_3	$\frac{P_2}{P_2+P_3}$	P_4	P_5	$\frac{P_4}{P_4+P_5}$
1	4,8990	,9798	,1211	,0270	,8175	,8247	,0230	,9763
1,5	4,7697	,9539	,2486	,0255	,9070	,6973	,0245	,9660
2	4,5826	,9165	,4139	,0257	,9415	,5319	,0243	,9563
2,5	4,3301	,8660	,5884	,0281	,9544	,3574	,0219	,9423
3	4,0	,8	,7380	,0334	,9567	,2078	,0166	,9262
3,5	3,5707	,7141	,8425	,0403	,9543	,1033	,0097	,9142
4	3,0	,6	,9029	,0459	,9516	,0429	,0041	,9137
4,4	2,3749	,4750	,9280	,0486	,9503	,0178	,0014	,9251
4,6	1,9596	,3919	,9355	,0493	,9500	,0103	,0007	,9350
4,8	1,4	,2800	,9407	,0497	,9498	,0051	,0003	,9478

$P_2/(P_2 + P_3)$ increases, then decreases, as δ_1 increases. $P_4/(P_4 + P_5)$ decreases, then increases, as δ_1 increases. It seems safest to have δ_1 large and δ_2 small.

TABLE 5

The effect of variation in δ_1 while $\delta_2 = 0$ $(f = 10; \delta_2 = 0; \lambda = \delta_1^2; \rho = 0)$

δ_1	P_1	P_2	P_3	$\frac{P_2}{P_2+P_3}$	P_4	P_5	$\frac{P_4}{P_4+P_5}$
1	,7573	,1351	,0130	,9122	,0576	,0370	,6089
2	,4695	,4093	,0303	,9310	,0712	,0197	,7836
3	,1795	,7272	,0442	,9427	,0432	,0058	,8819
4	,0380	,8997	,0491	,9482	,0123	,0009	,9355
5	,0042	,9442	,0499	,9498	,0017	,0001	,9670

Both conditional coverage probabilities increase with δ_1 .

5. Discussion

In most practical applications a confidence set which has infinite length is undesirable, which means that for the present problem one would try to obtain an interval as confidence set. The quantities at one's disposal when planning the experiment are v_{11} , v_{12} and v_{22} which in turn influence δ_1 and δ_2 . Thus, for example, δ_1^2 increases as v_{22} decreases. From the properties of the Student t-test it follows that $P_2 + P_3 = P\{T_1^2 > t^2\}$ tends to 1 as δ_1^2 increases, which means that $P_2 \rightarrow 1 - \gamma$ and $P_3 \rightarrow \gamma$ as δ_1^2 increases. Hence it follows that $P_2/(P_2 + P_3) \rightarrow \gamma$ as δ_1^2 increases. Thus one can expect that the conditional probability of coverage, given that the confidence set is an interval, tends to $1 - \gamma$ as δ_1^2 increases. From the tables presented here, it appears that one can expect the conditional coverage probability to be near $1 - \gamma$ if the probability of obtaining an interval is fairly high (larger than about 0.8 when $\gamma = 0.05$). The length of the interval is a decreasing function of T_1^2 , which in turn increases stochastically with δ_1^2 . Thus, by choosing v_{22} as small as possible, one ensures that the interval will be as short as possible.

In the case of one regression line where the values of the independent variable are $x_1, \dots, x_n$, v_{22} is given by $v_{22} = 1/\Sigma(x_i - \bar{x})^2$ where $\bar{x} = \Sigma x_i/n$. Thus δ_1^2 is increased by increasing the number of observations not at $\bar{x}$ and by taking the observations as far away from $\bar{x}$ as possible. The same holds for each of two regression lines when the purpose is to estimate their intersection point; the optimum allocation is to assign equal numbers of observations to the two lines. In the problem of estimating the stationary point of a quadratic regression, the optimum design as far as δ_1^2 is concerned would be to take one quarter of the observations at each of the two extremes and the other half in the middle.

The effect of δ_2^2 is also to increase the conditional coverage probability $P_2/(P_2 + P_3)$ if δ_1^2 is not decreased at the same time. The length of the confidence interval is an increasing function of T_2^2 , however, which means that the length of the interval increases stochastically with δ_2^2 . Thus the increase in coverage probability is due to an increase in length. In view of the fact that one can obtain the desired conditional coverage probability regardless of δ_2^2 , it would seem desirable to plan for a small δ_2^2 . From the definition of δ_2 it follows that δ_2^2 increases as v_{22}/v increases. In the case of one regression line $v_{22}/v = n$, the number of observations. However, one can make δ_2^2 small by choosing v_{12}/v_{22} as near as possible to θ . In regression applications this means that one should try to perform the experiment as near as possible to the solution. In the case of one regression line, for example, $v_{12}/v_{22} = \bar{y}$, the means of the observations. In the case of quadratic regression v_{12}/v_{22} is the predicted regression slope at $\bar{x}$ provided the x_i are symmetric about $\bar{x}$.

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